

# BIG DATA

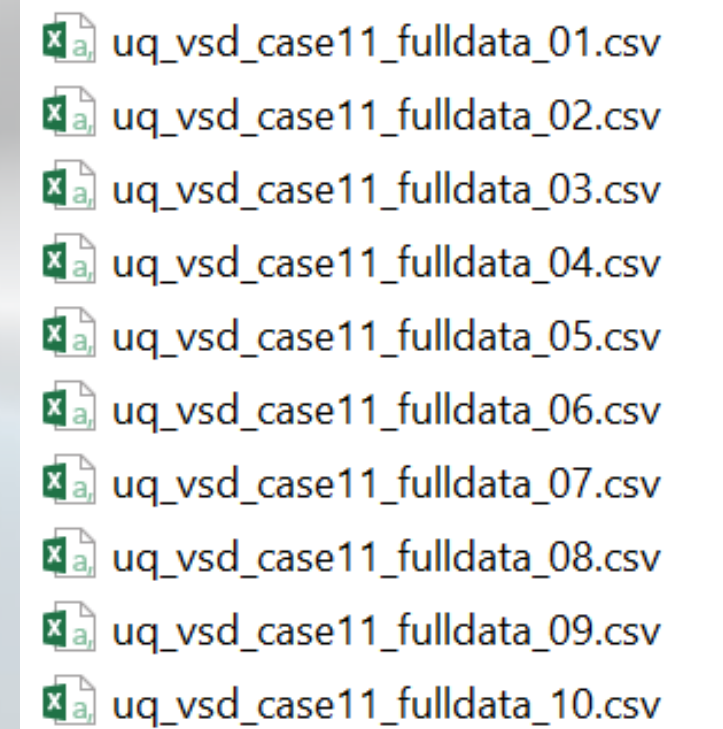
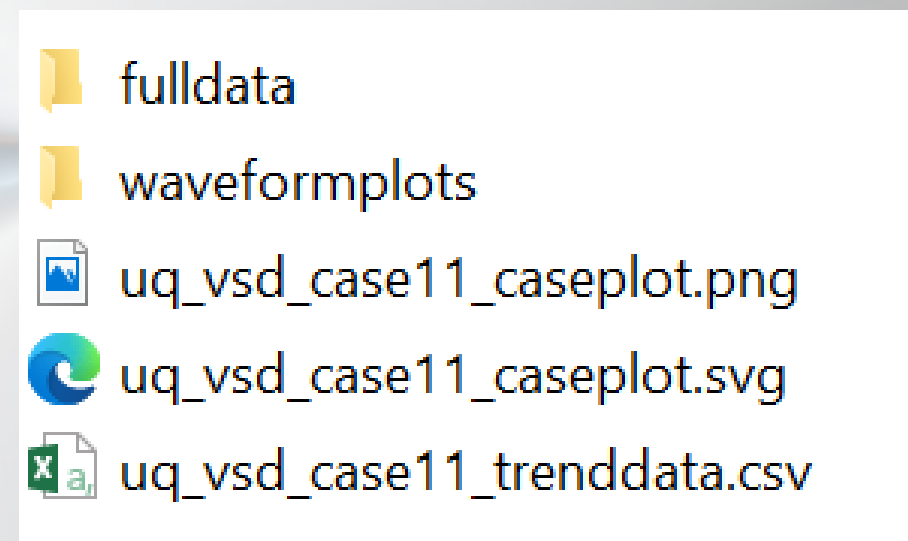
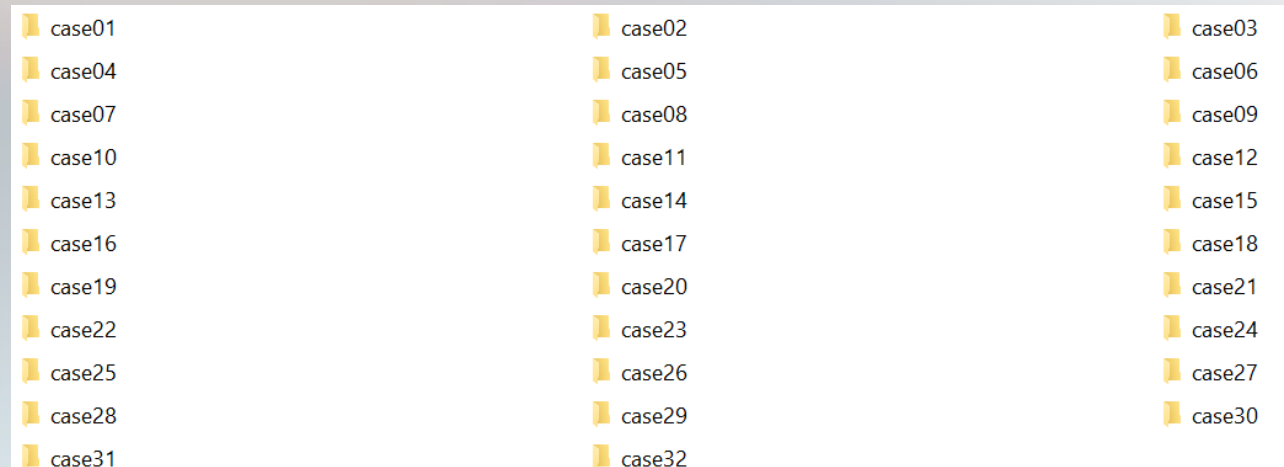


# PROJECT 1

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# STEP 1: Preprocessing and reformatting data

Structure of the downloaded files (4.14GB):



Run **prepare\_data.py** to merge and clean the folders into a unified dataset.

# STEP 2: Docker-compose file

namenode:

- ports: **9870** for HDFS Web UI and **9000** for external access to HDFS.
- volumes: - **./dataset\_ready\_for\_hdfs:/data/incoming** - connecting my data folder with the namenode container folder.
- **hadoop\_namenode:/hadoop/dfs/name** - for Metadata Persistence, so as not to lose the data after stopping the container, but for the FileSystem Image.
- environment:
- **CORE\_CONF\_fs\_defaultFS=hdfs://namenode:8020** - default port 8020 for internal communication inside the Docker network.

# STEP 2: Docker-compose file

datanode:

- volumes:

- **hadoop\_datanode:/hadoop/dfs/name** - for Metadata Persistence, so as not to lose the data after stopping the container, but for the physical data.

- environment:

- **SERVICE\_PRECONDITION=namenode:9870** - to prevent starting before the namenode.

- **CORE\_CONF\_fs\_defaultFS=hdfs://namenode:8020** - pointing towards the namenode.

# STEP 2: Docker-compose file

spark-master:

- environment:
  - **SPARK\_DRIVER\_MEMORY=2G** - added after application crashes.
- ports: **8080** for Web UI and **7077** as Master RPC Port between workers and the master.
- volumes: `./app` - creation of the `/app` folder inside the Spark container for code synchronization



# STEP 2: Docker-compose file

spark-worker:

- environment:
  - **SPARK\_WORKER\_CORES=4** - speeding up the processing time.
  - **SPARK\_WORKER\_MEMORY=8G** - giving more RAM to the worker.

Added because of errors caused by the quantity of the data and the groupby function (Shuffle) in **task2**.

- volumes: the same as spark-master, so it can have access to the same files.

# STEP 2: Docker-compose file

volumes:

hadoop\_datanode:

hadoop\_namenode:

★ Creating a permanent space on the disk, which is not erased after stopping the containers.

*In this specific Hadoop environment, the HDFS metadata and data blocks are highly sensitive to container lifecycles. Even with named volumes, restarting the cluster with **down** can trigger a re-initialization (formatting) of the NameNode, leading to a Cluster ID mismatch. To ensure Data Consistency and avoid re-uploading the 21M records, the system should be managed using **stop** and **start** instead of **down** and **up**.*

# STEP 2: Docker-compose file

- *docker compose up -d*

|                          |   |   |              |              |                           |                                                                   |       |           |   |   |   |
|--------------------------|---|---|--------------|--------------|---------------------------|-------------------------------------------------------------------|-------|-----------|---|---|---|
| <input type="checkbox"/> | ▼ | ● | project1     | -            | -                         | -                                                                 | 0.64% | 2 hours a | ■ | ⋮ | 🗑 |
| <input type="checkbox"/> |   | ● | namenode     | 831ee0240827 | <a href="#">bde2020/h</a> | <a href="#">9000:9000</a> ↗<br><a href="#">Show all ports (2)</a> | 0.25% | 2 hours a | ■ | ⋮ | 🗑 |
| <input type="checkbox"/> |   | ● | datanode     | f91c856e6f79 | <a href="#">bde2020/h</a> |                                                                   | 0.24% | 2 hours a | ■ | ⋮ | 🗑 |
| <input type="checkbox"/> |   | ● | spark-master | cc67acbf2f61 | <a href="#">bde2020/s</a> | <a href="#">7077:7077</a> ↗<br><a href="#">Show all ports (2)</a> | 0.08% | 2 hours a | ■ | ⋮ | 🗑 |
| <input type="checkbox"/> |   | ● | spark-worker | 8dba8242feb8 | <a href="#">bde2020/s</a> | <a href="#">8081:8081</a> ↗                                       | 0.07% | 2 hours a | ■ | ⋮ | 🗑 |



# STEP 3: SENDING FILES TO HDFS

- Making a space to receive the files:

```
docker exec -it namenode hdfs dfs -mkdir -p /user/root
```

- Copying files from disk to HDFS:

```
docker exec -it namenode hdfs dfs -put /data/incoming /user/root/dataset
```

You can refresh the **localhost:9870** page while waiting for the log display to end, and follow the uploading at **DFS Used**.

# STEP 3: SENDING FILES TO HDFS

## Overview 'namenode:8020' (active)

|                |                                                                    |
|----------------|--------------------------------------------------------------------|
| Started:       | Sat Feb 14 13:31:02 +0100 2026                                     |
| Version:       | 3.2.1, rb3cbbb467e22ea829b3808f4b7b01d07e0bf3842                   |
| Compiled:      | Tue Sep 10 17:56:00 +0200 2019 by rohithsharmaks from branch-3.2.1 |
| Cluster ID:    | CID-da2cb362-6db1-43ea-b417-2041459a6a8b                           |
| Block Pool ID: | BP-1597449827-172.20.0.4-1771072261304                             |

## NameNode Storage

| Storage Directory | Type            | State  |
|-------------------|-----------------|--------|
| /hadoop/dfs/name  | IMAGE_AND_EDITS | Active |

# Summary

Security is off.

Safemode is off.

369 files and directories, 365 blocks (365 replicated blocks, 0 erasure coded block groups) = 734 total filesystem object(s).

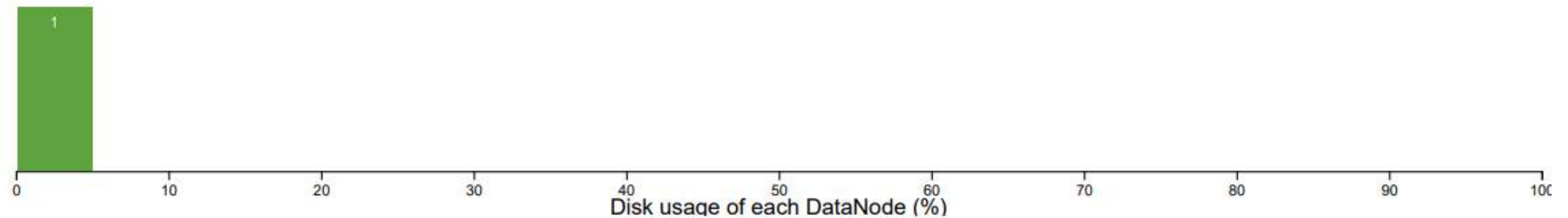
Heap Memory used 61.24 MB of 411.5 MB Heap Memory. Max Heap Memory is 1.69 GB.

Non Heap Memory used 69.53 MB of 71.19 MB Committed Non Heap Memory. Max Non Heap Memory is <unbounded>.

|                                                        |                                          |
|--------------------------------------------------------|------------------------------------------|
| Configured Capacity:                                   | 1006.85 GB                               |
| Configured Remote Capacity:                            | 0 B                                      |
| DFS Used:                                              | 4.14 GB (0.41%)                          |
| Non DFS Used:                                          | 9.61 GB                                  |
| DFS Remaining:                                         | 941.89 GB (93.55%)                       |
| Block Pool Used:                                       | 4.14 GB (0.41%)                          |
| DataNodes usages% (Min/Median/Max/stdDev):             | 0.41% / 0.41% / 0.41% / 0.00%            |
| Live Nodes                                             | 1 (Decommissioned: 0, In Maintenance: 0) |
| Dead Nodes                                             | 0 (Decommissioned: 0, In Maintenance: 0) |
| Decommissioning Nodes                                  | 0                                        |
| Entering Maintenance Nodes                             | 0                                        |
| Total Datanode Volume Failures                         | 0 (0 B)                                  |
| Number of Under-Replicated Blocks                      | 365                                      |
| Number of Blocks Pending Deletion (including replicas) | 0                                        |
| Block Deletion Start Time                              | Sat Feb 14 13:31:02 +0100 2026           |
| Last Checkpoint Time                                   | Sat Feb 14 13:31:01 +0100 2026           |
| Enabled Erasure Coding Policies                        | RS-6-3-1024k                             |

# STEP 3: SENDING FILES TO HDFS

Datanode usage histogram



## In operation

Show  entries

Search:

| Node                                     | Http Address                                           | Last contact | Last Block Report | Capacity   | Blocks | Block pool used    | Version |
|------------------------------------------|--------------------------------------------------------|--------------|-------------------|------------|--------|--------------------|---------|
| ✓ f91c856e6f79:9866<br>(172.20.0.2:9866) | http://f91c856e6f79:9864<br>(http://f91c856e6f79:9864) | 1s           | 37m               | 1006.85 GB | 365    | 4.14 GB<br>(0.41%) | 3.2.1   |

# STEP 3: SENDING FILES TO HDFS

## Browse Directory

Show 25 entries

Search:

| <input type="checkbox"/> | Permission | Owner | Group      | Size     | Last Modified | Replication | Block Size | Name                          |  |
|--------------------------|------------|-------|------------|----------|---------------|-------------|------------|-------------------------------|--|
| <input type="checkbox"/> | -rw-r--r-- | root  | supergroup | 10.75 MB | Feb 14 13:33  | 3           | 128 MB     | uq_vsd_case01_fulldata_01.csv |  |
| <input type="checkbox"/> | -rw-r--r-- | root  | supergroup | 10.24 MB | Feb 14 13:33  | 3           | 128 MB     | uq_vsd_case01_fulldata_02.csv |  |
| <input type="checkbox"/> | -rw-r--r-- | root  | supergroup | 10.19 MB | Feb 14 13:33  | 3           | 128 MB     | uq_vsd_case01_fulldata_03.csv |  |
| <input type="checkbox"/> | -rw-r--r-- | root  | supergroup | 10.23 MB | Feb 14 13:33  | 3           | 128 MB     | uq_vsd_case01_fulldata_04.csv |  |
| <input type="checkbox"/> | -rw-r--r-- | root  | supergroup | 10.25 MB | Feb 14 13:33  | 3           | 128 MB     | uq_vsd_case01_fulldata_05.csv |  |



# STEP 4 (OPTIONAL): CHECKING FUNCTIONALITY

Run the *spark-analysis.py* file to count the number of samples:

***docker exec -it spark-master /spark/bin/spark-submit /app/spark\_analysis.py***

Output: **20.909.657** samples

# STEP 5: PROJECT PYTHON CODE

---

- Use of argparse lib to accept parameters at runtime:

```
parser = argparse.ArgumentParser(description='BDS Project 1 - Spark Batch Processing')
parser.add_argument('--task', type=int, required=True, help='1 for filtering, 2 for stats')
parser.add_argument('--col', type=str, help='Filter (task 1) or stat (task 2) attribute')
parser.add_argument('--val', type=float, help='Granična vrednost za filtriranje (task 1)')
parser.add_argument('--group_by', type=str, default='case_name', help='Grouping attribute (task 2)')
parser.add_argument('--sort_by', type=str, default='RelativeTimeMilliseconds', help='Sorting column (T1)')

args = parser.parse_args()
```



# STEP 5: PROJECT PYTHON CODE

- Opening the Spark session, the address matches the service name and port in the YAML file:

```
spark = SparkSession.builder \  
    .appName("BDS_Final_Project") \  
    .master("spark://spark-master:7077") \  
    .getOrCreate()
```



# STEP 5: PROJECT PYTHON CODE

- Reading data from HDFS, combining it into a dataframe with a new column `case_name`, reformatting the `case_name` column values:

```
hdfs_path = "hdfs://namenode:8020/user/root/dataset/*.csv"
df = spark.read.csv(hdfs_path, header=True, inferSchema=True) \
    .withColumn("case_name", F.input_file_name())

# Cleaning 'case_name' so that only the name of the file is left, and not the whole path
df = df.withColumn("case_name", F.element_at(F.split(F.col("case_name"), "/"), -1))
```



# STEP 5: PROJECT PYTHON CODE

- Task 1 and Task 2 implementation:

```
if args.task == 1:
    print(f"--- TASK 1: Filtering {args.col} > {args.val}, Sorted by: {args.sort_by} ---")
    result = df.filter(F.col(args.col) > args.val).orderBy(args.sort_by)
    result.select("case_name", "Time", args.col).show(20)
    print(f"Total number of samples: {result.count()}")

elif args.task == 2:
    print(f"--- TASK 2: Stats for {args.col} grouped by {args.group_by} ---")
    stats = df.groupBy(args.group_by).agg(
        F.min(args.col).alias("Min"),
        F.max(args.col).alias("Max"),
        F.avg(args.col).alias("Mean"),
        F.stddev(args.col).alias("Std_Dev"),
        F.variance(args.col).alias("Variance"),
        F.percentile_approx(args.col, 0.5).alias("Median"),
        F.count("*").alias("Samples count")
    )
    stats.show()
```





# STEP 6: RUN THE SCRIPT

Run (for Task 1):

```
docker exec -it spark-master /spark/bin/spark-submit --driver-memory 2G --executor-memory 4G  
/app/spark_final.py --task 1 --col SpO2 --val 90
```

```
+-----+-----+-----+
|          case_name|          Time|SpO2|
+-----+-----+-----+
|uq_vsd_case27_ful...|00:00:00_000| 95|
|uq_vsd_case16_ful...|00:00:00_000| 98|
|uq_vsd_case23_ful...|00:00:00_000| 99|
|uq_vsd_case25_ful...|00:00:00_000| 98|
|uq_vsd_case13_ful...|00:00:00_000| 97|
|uq_vsd_case08_ful...|00:00:00_000| 99|
|uq_vsd_case21_ful...|00:00:00_000| 100|
|uq_vsd_case15_ful...|00:00:00_000| 97|
|uq_vsd_case09_ful...|00:00:00_000| 100|
|uq_vsd_case07_ful...|00:00:00_000| 99|
|uq_vsd_case05_ful...|00:00:00_000| 99|
|uq_vsd_case31_ful...|00:00:00_000| 100|
|uq_vsd_case28_ful...|00:00:00_000| 95|
|uq_vsd_case01_ful...|00:00:00_000| 100|
|uq_vsd_case25_ful...|00:00:00_010| 98|
|uq_vsd_case28_ful...|00:00:00_010| 95|
|uq_vsd_case15_ful...|00:00:00_010| 97|
|uq_vsd_case08_ful...|00:00:00_010| 99|
|uq_vsd_case09_ful...|00:00:00_010| 100|
|uq_vsd_case05_ful...|00:00:00_010| 99|
+-----+-----+-----+
only showing top 20 rows
```



## STEP 6: RUN THE SCRIPT

## Run (for Task 2):

```
docker exec -it spark-master /spark/bin/spark-submit --driver-memory 2G --executor-memory 4G /app/spark_final.py --task 2 --col HR --group_by SpO2
```

| SpO2 | Min | Max | Mean              | Std_Dev            | Variance           | Median | Samples count |
|------|-----|-----|-------------------|--------------------|--------------------|--------|---------------|
| 85   | 69  | 95  | 85.63532401524778 | 9.877773259720685  | 97.57040457045301  | 91     | 1094          |
| 78   | 61  | 96  | 78.38235294117646 | 16.392811556381027 | 268.72427072301934 | 94     | 715           |
| 81   | 94  | 96  | 94.66449511400651 | 0.9435764890364262 | 0.890336590662309  | 94     | 307           |
| 76   | 60  | 95  | 75.33333333333333 | 14.625435580064726 | 213.90336590662326 | 71     | 615           |
| 91   | 57  | 101 | 78.9886345445814  | 11.23592222718436  | 126.24594829533557 | 80     | 9730          |
| 93   | 58  | 102 | 79.73931075343143 | 9.137055482233043  | 83.48578288540492  | 80     | 20971         |
| null | 55  | 112 | 75.76367107922636 | 14.071578536722242 | 198.0093225151421  | 71     | 403227        |
| 86   | 54  | 96  | 75.72727272727273 | 13.544698226847595 | 183.4588500563684  | 80     | 3174          |
| 94   | 57  | 118 | 78.94765946718212 | 11.97980380710562  | 143.51569925674232 | 78     | 64412         |
| 96   | 53  | 133 | 87.38446990952441 | 14.236130707620529 | 202.66741752445617 | 92     | 444251        |
| 92   | 63  | 98  | 76.93981978116284 | 9.525910542611289  | 90.7429716658329   | 78     | 10550         |
| 88   | 64  | 99  | 86.40642939150402 | 8.728404234284119  | 76.18504047706894  | 84     | 3893          |
| 100  | 0   | 165 | 73.7551021429644  | 12.686681003650214 | 160.9518748883792  | 72     | 12289467      |
| 84   | 62  | 120 | 94.27700503107428 | 14.312182922778538 | 204.8385800150736  | 95     | 3379          |
| 87   | 54  | 96  | 80.43275488069415 | 15.201803239035586 | 231.09482171835282 | 81     | 1230          |
| 97   | 52  | 135 | 89.51808515648591 | 19.47949393376205  | 379.4506839154725  | 92     | 786430        |
| 77   | 61  | 96  | 73.18811002661934 | 16.133383873175895 | 260.2860751992521  | 61     | 1127          |
| 82   | 63  | 98  | 90.52169197396964 | 11.974732090670965 | 143.39420864334502 | 96     | 922           |
| 80   | 64  | 94  | 78.92682926829268 | 15.03654085804344  | 226.09756097560975 | 64     | 205           |
| 73   | 95  | 95  | 95.0              | 0.0                | 0.0                | 95     | 511           |





# STEP 6: RUN THE SCRIPT

Run (app as a separate container):

```
docker run --rm -it --network project1_bigdata-network -v ${PWD}:/app bde2020/spark-master:3.1.1-hadoop3.2 /spark/bin/spark-submit --master spark://spark-master:7077 --driver-memory 2G --executor-memory 4G /app/spark_final.py --task 2 --col HR --group_by case_name
```

| case_name            | Min | Max | Mean               | Std_Dev            | Variance            | Median | Samples count |
|----------------------|-----|-----|--------------------|--------------------|---------------------|--------|---------------|
| uq_vsd_case27_ful... | 67  | 71  | 68.56643333333334  | 1.0831925809038478 | 1.1733061673251388  | 68     | 60000         |
| uq_vsd_case22_ful... | 85  | 110 | 97.90175472425761  | 5.7506543338107425 | 33.07002526697627   | 98     | 60000         |
| uq_vsd_case21_ful... | 80  | 92  | 85.86305           | 3.630401404951478  | 13.179814361073666  | 85     | 60000         |
| uq_vsd_case05_ful... | 59  | 66  | 61.115516666666664 | 1.8288962921436505 | 3.344861647416793   | 60     | 60000         |
| uq_vsd_case21_ful... | 81  | 136 | 111.0306           | 14.706041891044046 | 216.26766810114236  | 117    | 60000         |
| uq_vsd_case01_ful... | 51  | 56  | 53.228466666666667 | 1.0900716122458087 | 1.1882561198241768  | 53     | 60000         |
| uq_vsd_case17_ful... | 72  | 90  | 77.24006678359417  | 3.0525809183127888 | 9.31825026284735    | 77     | 60000         |
| uq_vsd_case31_ful... | 70  | 86  | 76.40365           | 1.838134830880067  | 3.378739656494492   | 76     | 60000         |
| uq_vsd_case32_ful... | 62  | 78  | 66.707216666666667 | 3.5978572784372824 | 12.94457699600413   | 66     | 60000         |
| uq_vsd_case31_ful... | 68  | 87  | 79.79916666666666  | 4.5678967151447205 | 20.865680400229927  | 81     | 60000         |
| uq_vsd_case32_ful... | 66  | 77  | 70.128             | 2.4558670949488017 | 6.031283188052266   | 70     | 60000         |
| uq_vsd_case04_ful... | 125 | 128 | 126.06986666666667 | 0.7618584663139701 | 0.5804283226942747  | 126    | 60000         |
| uq_vsd_case27_ful... | 68  | 90  | 73.12357220503387  | 2.2846120463172066 | 5.219452202177694   | 73     | 60000         |
| uq_vsd_case27_ful... | 76  | 94  | 84.53896666666667  | 2.0082286096435755 | 4.032982148590968   | 85     | 60000         |
| uq_vsd_case32_ful... | 67  | 97  | 78.86293333333333  | 5.746842306765533  | 33.026196498830195  | 80     | 60000         |
| uq_vsd_case03_ful... | 61  | 69  | 64.32716666666667  | 1.7735128214980067 | 3.145347728017821   | 64     | 60000         |
| uq_vsd_case31_ful... | 72  | 82  | 76.80461666666666  | 2.8194103114477773 | 7.9490745042980535  | 77     | 60000         |
| uq_vsd_case07_ful... | 58  | 66  | 60.66481666666667  | 2.5132861774667425 | 6.316607409845391   | 59     | 60000         |
| uq_vsd_case03_ful... | 58  | 79  | 70.76603333333334  | 3.597254777778937  | 12.94024193625339   | 71     | 60000         |
| uq_vsd_case01_ful... | 56  | 59  | 57.208216666666665 | 0.6380720481689658 | 0.40713593865453906 | 57     | 60000         |

only showing top 20 rows



# STEP 6: RUN THE SCRIPT

Separate container:

| <input type="checkbox"/> |     | Name            | Container ID | Image                     | Port(s)                                                           | CPU (%) | Last star | Actions                    |
|--------------------------|-----|-----------------|--------------|---------------------------|-------------------------------------------------------------------|---------|-----------|----------------------------|
| <input type="checkbox"/> | ●   | thirsty_ellis   | 3c2d73866d3b | <a href="#">bde2020/s</a> |                                                                   | 129.48% | 2 second  | <input type="checkbox"/> ⋮ |
| <input type="checkbox"/> | ▼ ● | <u>project1</u> | -            | -                         | -                                                                 | 0.64%   | 2 hours a | <input type="checkbox"/> ⋮ |
| <input type="checkbox"/> | ●   | namenode        | 831ee0240827 | <a href="#">bde2020/h</a> | <a href="#">9000:9000</a> ↗<br><a href="#">Show all ports (2)</a> | 0.25%   | 2 hours a | <input type="checkbox"/> ⋮ |
| <input type="checkbox"/> | ●   | datanode        | f91c856e6f79 | <a href="#">bde2020/h</a> |                                                                   | 0.24%   | 2 hours a | <input type="checkbox"/> ⋮ |
| <input type="checkbox"/> | ●   | spark-master    | cc67acbf2f61 | <a href="#">bde2020/s</a> | <a href="#">7077:7077</a> ↗<br><a href="#">Show all ports (2)</a> | 0.08%   | 2 hours a | <input type="checkbox"/> ⋮ |
| <input type="checkbox"/> | ●   | spark-worker    | 8dba8242feb8 | <a href="#">bde2020/s</a> | <a href="#">8081:8081</a> ↗                                       | 0.07%   | 2 hours a | <input type="checkbox"/> ⋮ |







## Spark Master at spark://cc67acbf2f61:7077

URL: spark://cc67acbf2f61:7077

Alive Workers: 1

Cores in use: 4 Total, 4 Used

Memory in use: 8.0 GiB Total, 4.0 GiB Used

Resources in use:

Applications: 1 [Running](#), 3 [Completed](#)

Drivers: 0 Running, 0 Completed

Status: ALIVE

### Workers (1)

| Worker Id                                              | Address          | State | Cores      | Memory                 | Resources |
|--------------------------------------------------------|------------------|-------|------------|------------------------|-----------|
| <a href="#">worker-20260214123102-172.20.0.5-33725</a> | 172.20.0.5:33725 | ALIVE | 4 (4 Used) | 8.0 GiB (4.0 GiB Used) |           |

### Running Applications (1)

| Application ID                                    | Name                              | Cores | Memory per Executor | Resources Per Executor | Submitted Time      | User | State   | Duration |
|---------------------------------------------------|-----------------------------------|-------|---------------------|------------------------|---------------------|------|---------|----------|
| <a href="#">app-20260214141902-0003</a><br>(kill) | <a href="#">BDS_Final_Project</a> | 4     | 4.0 GiB             |                        | 2026/02/14 14:19:02 | root | RUNNING | 29 s     |

### Completed Applications (3)

| Application ID                          | Name              | Cores | Memory per Executor | Resources Per Executor | Submitted Time      | User | State    | Duration |
|-----------------------------------------|-------------------|-------|---------------------|------------------------|---------------------|------|----------|----------|
| <a href="#">app-20260214141449-0002</a> | BDS_Final_Project | 4     | 4.0 GiB             |                        | 2026/02/14 14:14:49 | root | FINISHED | 2.9 min  |
| <a href="#">app-20260214125715-0001</a> | BDS_Final_Project | 4     | 1024.0 MiB          |                        | 2026/02/14 12:57:15 | root | FINISHED | 2.5 min  |
| <a href="#">app-20260214124935-0000</a> | BDS_Final_Project | 4     | 1024.0 MiB          |                        | 2026/02/14 12:49:35 | root | FINISHED | 2.6 min  |



## Application: BDS\_Final\_Project

**ID:** app-20260214141902-0003

**Name:** BDS\_Final\_Project

**User:** root

**Cores:** Unlimited (4 granted)

**Executor Limit:** Unlimited (1 granted)

**Executor Memory:** 4.0 GiB

**Executor Resources:**

**Submit Date:** 2026/02/14 14:19:02

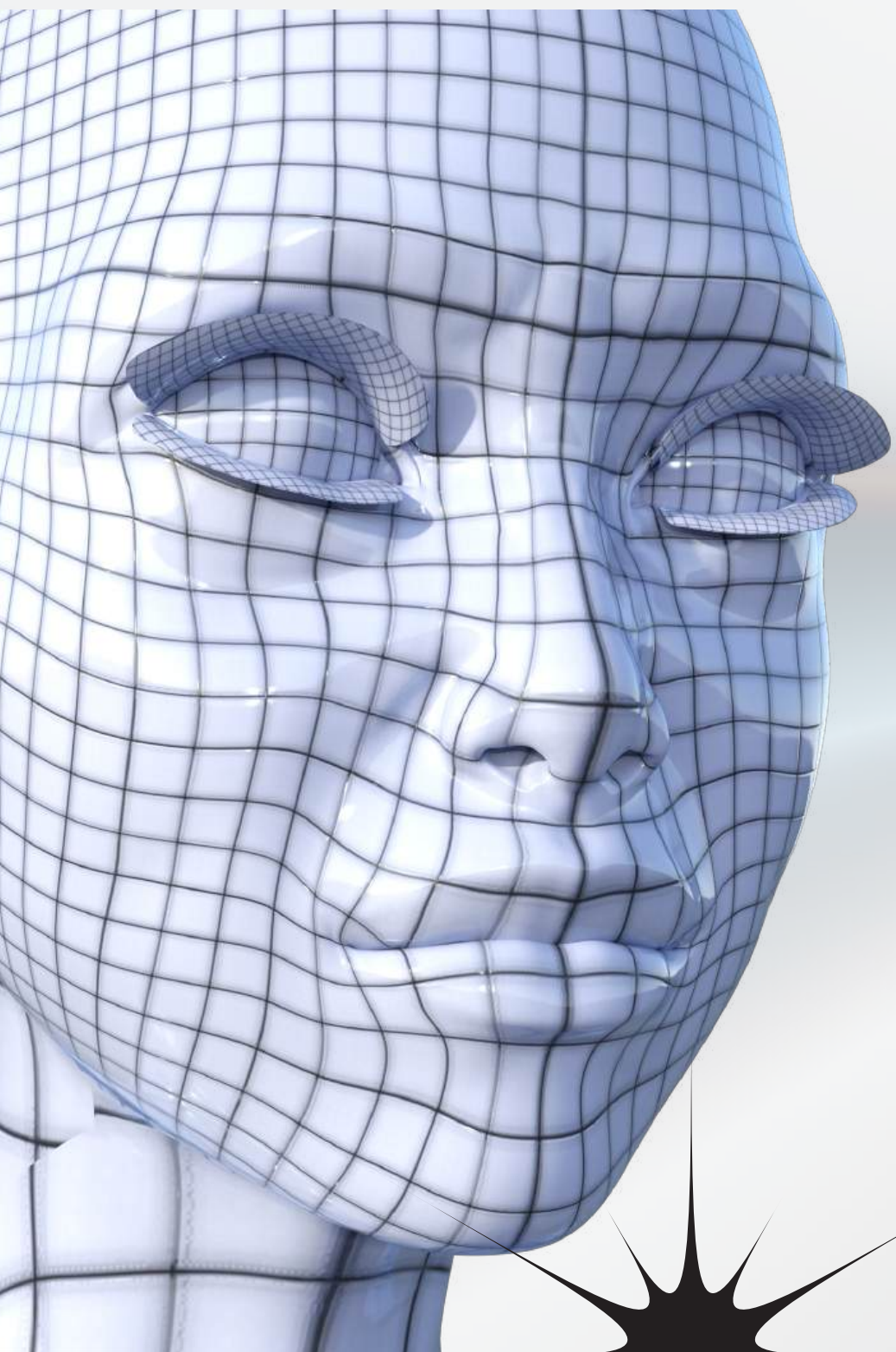
**State:** RUNNING

[Application Detail UI](#)

### ▼ Executor Summary (1)

| ExecutorID | Worker                                                 | Cores | Memory | Resources | State   | Logs                                          |
|------------|--------------------------------------------------------|-------|--------|-----------|---------|-----------------------------------------------|
| 0          | <a href="#">worker-20260214123102-172.20.0.5-33725</a> | 4     | 4096   |           | RUNNING | <a href="#">stdout</a> <a href="#">stderr</a> |





# Thank You

Stay Safe, Stay Smart

